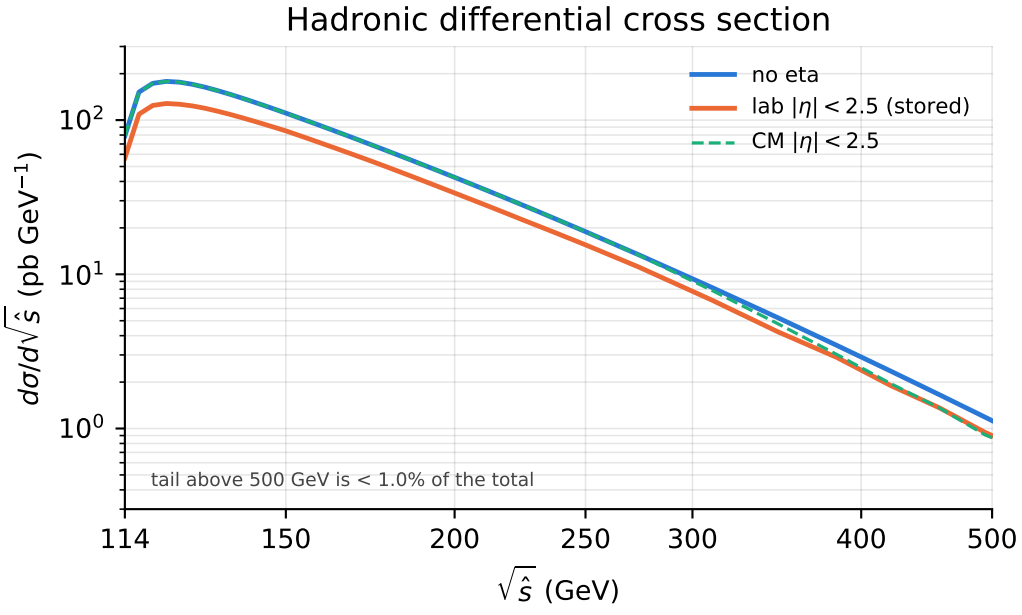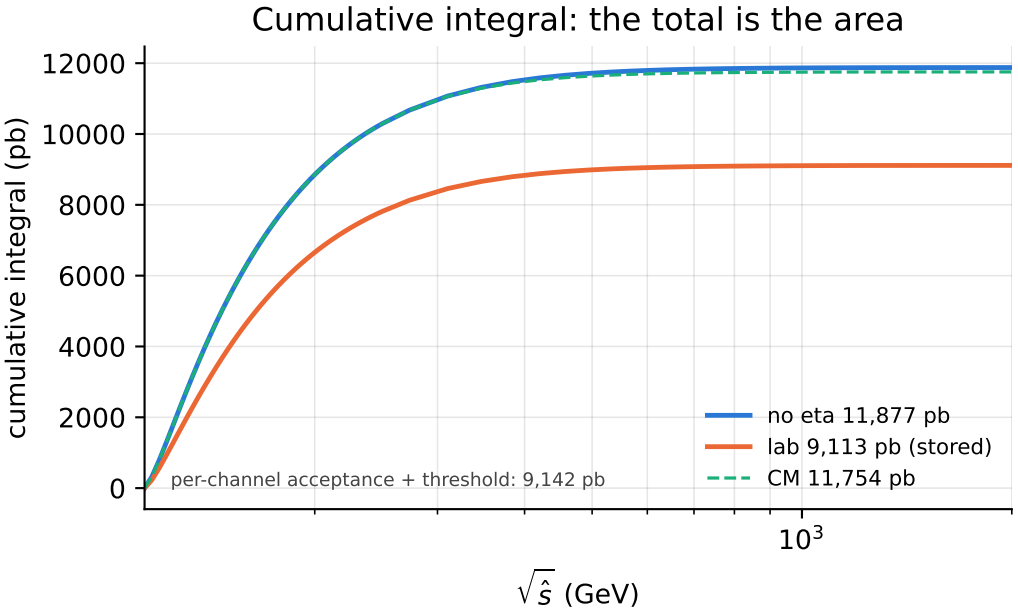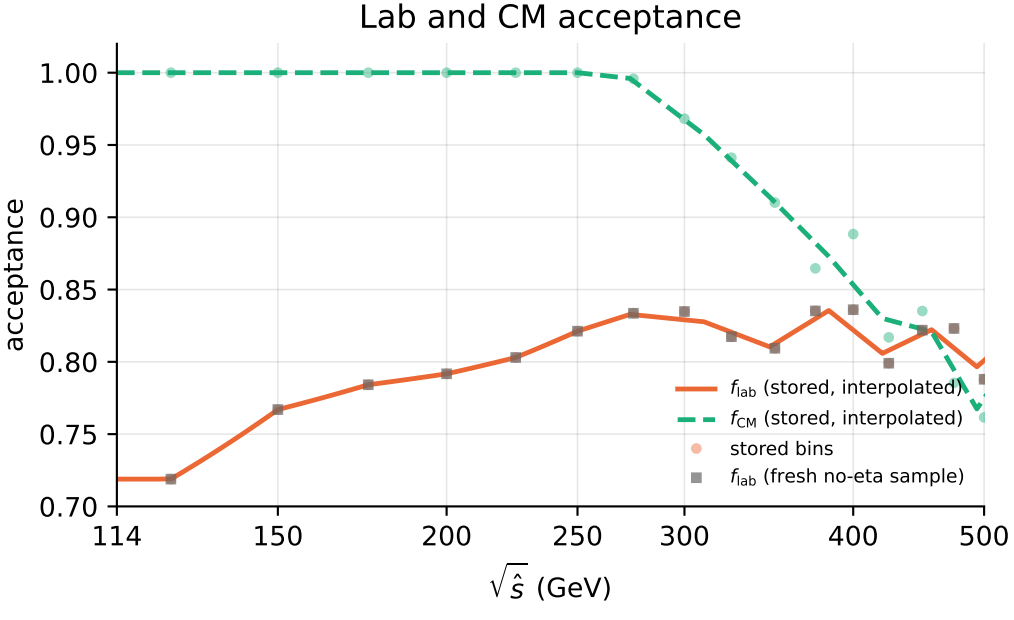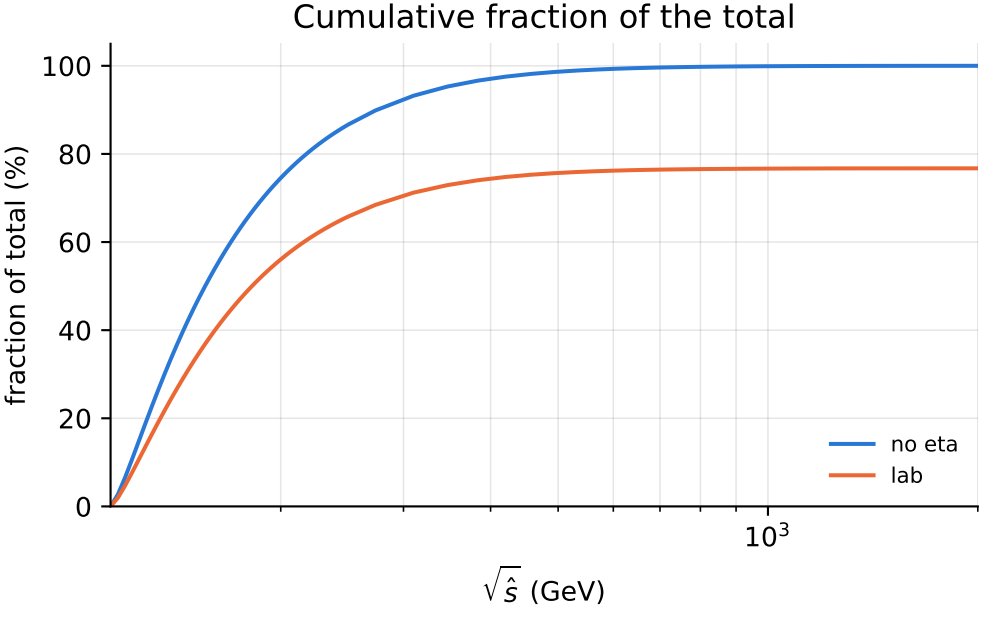From the partonic scan to the lab-frame hadronic total